

**Theorem 2.3.** *The following are all logical equivalences.*

- (a)  $(\phi \wedge \psi) \equiv (\psi \wedge \phi)$  (commutativity of  $\wedge$ )
- (b)  $(\phi \vee \psi) \equiv (\psi \vee \phi)$  (commutativity of  $\vee$ )
- (c)  $(\phi \wedge \phi) \equiv \phi$  (idempotence of  $\wedge$ )
- (d)  $(\phi \vee \phi) \equiv \phi$  (idempotence of  $\vee$ )
- (e)  $(\phi \wedge (\psi \wedge \theta)) \equiv ((\phi \wedge \psi) \wedge \theta)$  (associativity of  $\wedge$ )
- (f)  $(\phi \vee (\psi \vee \theta)) \equiv ((\phi \vee \psi) \vee \theta)$  (associativity of  $\vee$ )
- (g)  $\neg\neg\phi \equiv \phi$  (law of double negation)
- (h)  $\neg(\phi \wedge \psi) \equiv (\neg\phi \vee \neg\psi)$  (De Morgan's Law)
- (i)  $\neg(\phi \vee \psi) \equiv (\neg\phi \wedge \neg\psi)$  (De Morgan's Law)
- (j)  $(\phi \wedge (\psi \vee \theta)) \equiv ((\phi \wedge \psi) \vee (\phi \wedge \theta))$  (distributivity of  $\wedge$  over  $\vee$ )
- (k)  $(\phi \vee (\psi \wedge \theta)) \equiv ((\phi \vee \psi) \wedge (\phi \vee \theta))$  (distributivity of  $\vee$  over  $\wedge$ )
- (l)  $(\phi \wedge (\psi \vee \phi)) \equiv \phi$  (absorption law for  $\wedge$ )
- (m)  $(\phi \vee (\psi \wedge \phi)) \equiv \phi$  (absorption law for  $\vee$ )

*Proof of (a).* Let  $v$  be any truth assignment. WLOG if  $v(\phi) = F$  and  $\psi$ 's truth value is free, then  $v((\phi \wedge \psi)) = F = v((\psi \wedge \phi))$ . If  $v(\phi) = T = v(\psi)$ , then  $v((\phi \wedge \psi)) = T = v((\psi \wedge \phi))$ . Thus,  $(\phi \wedge \psi) \equiv (\psi \wedge \phi)$ .  $\square$

*Proof of (b).* Let  $v$  be any truth assignment. WLOG if  $v(\phi) = T$  and  $\psi$ 's truth value is free, then  $v((\phi \vee \psi)) = T = v((\psi \vee \phi))$ . If  $v(\phi) = F = v(\psi)$ , then  $v((\phi \vee \psi)) = F = v((\psi \vee \phi))$ . Thus,  $(\phi \vee \psi) \equiv (\psi \vee \phi)$ .  $\square$

*Proof of (c).* Let  $v$  be any truth assignment. If  $v(\phi) = F$  then  $v((\phi \wedge \phi)) = F$ . If  $v(\phi) = T$  then  $v((\phi \wedge \phi)) = T$ . Thus,  $(\phi \wedge \phi) \equiv \phi$ .  $\square$

*Proof of (d).* Let  $v$  be any truth assignment. If  $v(\phi) = F$  then  $v((\phi \vee \phi)) = F$ . If  $v(\phi) = T$  then  $v((\phi \vee \phi)) = T$ . Thus,  $(\phi \vee \phi) \equiv \phi$ .  $\square$

*Proof of (e).* Let  $v$  be any truth assignment.

( $\rightarrow$ ): If  $v((\phi \wedge (\psi \wedge \theta))) = T$ , then  $v(\phi) = T = v((\psi \wedge \theta))$  which gives  $v(\psi) = T = v(\theta)$ , so that  $v((\phi \wedge \psi)) = T$ , giving  $v(((\phi \wedge \psi) \wedge \theta)) = T$ .

( $\leftarrow$ ): Similarly, if  $v(((\phi \wedge \psi) \wedge \theta)) = T$ , then  $v(\theta) = T = v((\phi \wedge \psi))$  which gives  $v(\phi) = T = v(\psi)$ , so that  $v((\psi \wedge \theta)) = T$ , giving  $v((\phi \wedge (\psi \wedge \theta))) = T$ .

Since  $v$  takes only the values  $T$  and  $F$ , the contrapositive of the forwards direction is  $v(((\phi \wedge \psi) \wedge \theta)) = F \Rightarrow v((\phi \wedge (\psi \wedge \theta))) = F$  while the contrapositive of the backwards direction is  $v((\phi \wedge (\psi \wedge \theta))) = F \Rightarrow v(((\phi \wedge \psi) \wedge \theta)) = F$ .

So that for all truth assignments  $v$ ,  $v((\phi \wedge (\psi \wedge \theta))) = v(((\phi \wedge \psi) \wedge \theta))$  giving that  $(\phi \wedge (\psi \wedge \theta)) \equiv ((\phi \wedge \psi) \wedge \theta)$ .  $\square$

*Proof of (f).* Let  $v$  be any truth assignment.

( $\rightarrow$ ): If  $v((\phi \vee (\psi \vee \theta))) = F$ , then  $v(\phi) = F = v((\psi \vee \theta))$  which gives  $v(\psi) = F = v(\theta)$ , so that  $v((\phi \vee \psi)) = F$ , giving  $v(((\phi \vee \psi) \vee \theta)) = F$ .

( $\leftarrow$ ): Similarly, if  $v(((\phi \vee \psi) \vee \theta)) = F$ , then  $v(\theta) = F = v((\phi \vee \psi))$  which gives  $v(\phi) = F = v(\psi)$ , so that  $v((\psi \vee \theta)) = F$ , giving  $v((\phi \vee (\psi \vee \theta))) = F$ .

Since  $v$  takes only the values  $T$  and  $F$ , the contrapositive of the forwards direction is  $v(((\phi \vee \psi) \vee \theta)) = T \Rightarrow v((\phi \vee (\psi \vee \theta))) = T$  while the contrapositive of the backwards direction is  $v((\phi \vee (\psi \vee \theta))) = T \Rightarrow v(((\phi \vee \psi) \vee \theta)) = T$ .

So that for all truth assignments  $v$ ,  $v((\phi \vee (\psi \vee \theta))) = v(((\phi \vee \psi) \vee \theta))$  giving that  $(\phi \vee (\psi \vee \theta)) \equiv ((\phi \vee \psi) \vee \theta)$ .  $\square$

*Proof of (g).* Let  $v$  be any truth assignment. Let  $v(\phi) = T$ . Then  $v(\neg\phi) = F$ ,  $v(\neg\neg\phi) = T$  as desired. Let  $v(\phi) = F$ . Then  $v(\neg\phi) = T$ ,  $v(\neg\neg\phi) = F$  as desired. So that for all truth assignments  $v$ ,  $v(\phi) = v(\neg\neg\phi)$  giving that  $\phi \equiv \neg\neg\phi$ .  $\square$

*Proof of (h).* Let  $v$  be any truth assignment.

( $\rightarrow$ ): If  $v(\neg(\phi \wedge \psi)) = F$  then we know that  $v((\phi \wedge \psi)) = T$  so  $v(\phi) = v(\psi) = T$ . Therefore  $v(\neg\phi) = F = v(\neg\psi)$ , thus  $v((\neg\phi \vee \neg\psi)) = F$ .

( $\leftarrow$ ): Similarly, if  $v((\neg\phi \vee \neg\psi)) = F$  then we know that  $v(\neg\phi) = F = v(\neg\psi)$  and so  $v(\phi) = T = v(\psi)$ . Therefore  $v((\phi \wedge \psi)) = T$ , thus  $v(\neg(\phi \wedge \psi)) = F$ .

Since  $v$  takes only the values  $T$  and  $F$ , the contrapositive of the forwards direction is  $v((\neg\phi \vee \neg\psi)) = T \Rightarrow v(\neg(\phi \wedge \psi)) = T$  while the contrapositive of the backwards direction is  $v(\neg(\phi \wedge \psi)) = T \Rightarrow v((\neg\phi \vee \neg\psi)) = T$ .

So that for all truth assignments  $v$ ,  $v(\neg(\phi \wedge \psi)) = v((\neg\phi \vee \neg\psi))$  giving that  $\neg(\phi \wedge \psi) \equiv (\neg\phi \vee \neg\psi)$ .  $\square$

*Proof of (i).* Let  $v$  be any truth assignment.

( $\rightarrow$ ): If  $v(\neg(\phi \vee \psi)) = T$  then we know that  $v((\phi \vee \psi)) = F$  so  $v(\phi) = v(\psi) = F$ . Therefore  $v(\neg\phi) = T = v(\neg\psi)$ , thus  $v((\neg\phi \wedge \neg\psi)) = T$ .

( $\leftarrow$ ): Similarly, if  $v((\neg\phi \wedge \neg\psi)) = T$  then we know that  $v(\neg\phi) = T = v(\neg\psi)$  so  $v(\phi) = v(\psi) = F$ . Therefore,  $v((\phi \vee \psi)) = F$ , thus  $v(\neg(\phi \vee \psi)) = T$ .

Since  $v$  takes only the values  $T$  and  $F$ , the contrapositive of the forwards direction is  $v((\neg\phi \wedge \neg\psi)) = F \Rightarrow v(\neg(\phi \vee \psi)) = F$  while the contrapositive of the backwards direction is  $v(\neg(\phi \vee \psi)) = F \Rightarrow v((\neg\phi \wedge \neg\psi)) = F$ .

So that for all truth assignments  $v$ ,  $v(\neg(\phi \vee \psi)) = v((\neg\phi \wedge \neg\psi))$  giving that  $\neg(\phi \vee \psi) \equiv (\neg\phi \wedge \neg\psi)$ .  $\square$

*Proof of (j).* Let  $v$  be any truth assignment.

( $\rightarrow$ ): If  $v((\phi \wedge (\psi \vee \theta))) = T$  then we know that  $v(\phi) = T = v((\psi \vee \theta))$ . WLOG let  $v(\psi) = T$  and the truth value of  $\theta$  is free. Therefore we have  $v((\phi \wedge \psi)) = T$  and so,  $v(((\phi \wedge \psi) \vee (\phi \wedge \theta))) = T$ .

( $\leftarrow$ ): Similarly, let  $v(((\phi \wedge \psi) \vee (\phi \wedge \theta))) = T$ . WLOG, let  $v((\phi \wedge \psi)) = T$  and let  $v((\phi \wedge \theta))$  be free. From this we know that  $v(\phi) = T = v(\psi)$  and the truth value of  $\theta$  is free. Therefore,  $v((\psi \vee \theta)) = T$  and so we have  $v((\phi \wedge (\psi \vee \theta))) = T$ .

Since  $v$  takes only the values  $T$  and  $F$ , the contrapositive of the forwards direction is  $v(((\phi \wedge \psi) \vee (\phi \wedge \theta))) = F \Rightarrow v((\phi \wedge (\psi \vee \theta))) = F$  while the

contrapositive of the backwards direction is  $v((\phi \wedge (\psi \vee \theta))) = F \Rightarrow v(((\phi \wedge \psi) \vee (\phi \wedge \theta))) = F$ .

So that for all truth assignments  $v$ ,  $v((\phi \wedge (\psi \vee \theta))) = v(((\phi \wedge \psi) \vee (\phi \wedge \theta)))$  giving that  $(\phi \wedge (\psi \vee \theta)) \equiv ((\phi \wedge \psi) \vee (\phi \wedge \theta))$ .  $\square$

*Proof of (k).* Let  $v$  be any truth assignment.

( $\rightarrow$ ): If  $v((\phi \vee (\psi \wedge \theta))) = F$ , then we know that  $v(\phi) = F = v((\psi \wedge \theta))$ . WLOG, let  $v(\psi) = F$  and the truth value of  $\theta$  is free. Thus we have  $v((\phi \vee \psi)) = F$ . Therefore,  $v(((\phi \vee \psi) \wedge (\phi \vee \theta))) = F$ .

( $\leftarrow$ ): Similarly, let  $v(((\phi \vee \psi) \wedge (\phi \vee \theta))) = F$ . WLOG, let  $v((\phi \vee \psi)) = F$  and the truth value of  $(\phi \vee \theta)$  is free. Then  $v(\phi) = F = v(\psi)$ . Therefore,  $v((\psi \wedge \theta)) = F$ . Thus,  $v((\phi \vee (\psi \wedge \theta))) = F$ .

Since  $v$  takes only the values  $T$  and  $F$ , the contrapositive of the forwards direction is  $v(((\phi \vee \psi) \wedge (\phi \vee \theta))) = T \Rightarrow v((\phi \vee (\psi \wedge \theta))) = T$  while the contrapositive of the backwards direction is  $v((\phi \vee (\psi \wedge \theta))) = T \Rightarrow v(((\phi \vee \psi) \wedge (\phi \vee \theta))) = T$ .

So for all truth assignments  $v$ ,  $v((\phi \vee (\psi \wedge \theta))) = v(((\phi \vee \psi) \wedge (\phi \vee \theta)))$  giving that  $(\phi \vee (\psi \wedge \theta)) \equiv ((\phi \vee \psi) \wedge (\phi \vee \theta))$ .  $\square$

*Proof of (l).* Let  $v$  be any truth assignment.

If  $v(\phi) = T$ , then  $v((\psi \vee \phi)) = T$ , so that  $v((\phi \wedge (\psi \vee \phi))) = T$ .

If  $v(\phi) = F$ , then  $v((\phi \wedge (\psi \vee \phi))) = F$ .

So that for all truth assignments  $v$ ,  $v((\phi \wedge (\psi \vee \phi))) = v(\phi)$  giving that  $(\phi \wedge (\psi \vee \phi)) \equiv \phi$ .  $\square$

*Proof of (m).* Let  $v$  be any truth assignment.

If  $v(\phi) = T$ , then  $v((\phi \vee (\psi \wedge \phi))) = T$ .

If  $v(\phi) = F$ , then  $v((\psi \wedge \phi)) = F$ , so that  $v((\phi \vee (\psi \wedge \phi))) = F$ .

So that for all truth assignments  $v$ ,  $v((\phi \vee (\psi \wedge \phi))) = v(\phi)$  giving that  $(\phi \vee (\psi \wedge \phi)) \equiv \phi$ .  $\square$